

Doctor Anywhere: Harnessing the power of data to make remote healthcare more intuitive

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ABOUT DOCTOR



Doctor Anywhere empowers patient healthcare providers by running its multi-cloud environment and insights platform on Google Cloud and leveraging BigQuery for real-time analytics.

About Doctor Anywhere

Doctor Anywhere combines innovation and technology to improve healthcare access for consumers on a digital platform. The platform serves more than 1 million users online and offline in the region, with services including video consultation with medical professionals, an online health and wellness marketplace where users can book home-based health screenings and vaccinations with a mobile medical team, and an offline panel of more than 1,300 medical practitioners and specialists. During COVID-19, Doctor Anywhere partnered with major insurers

Google Cloud results

- Modernizes data warehouse by using BigQuery to speed up real-time analysis for 200 GB of data per month
- Google Cloud AI/ML technologies enable Doctor Anywhere to gain insights that help make its services more patient-centric
- Easily handles more than 70% increase in app traffic during

Enabling healthcare access for people in Asia

to offer free medical advisory to their policyholders so patients can video consult a licensed doctor without leaving their homes. Doctor Anywhere is a member of Singapore Ministry of Health's regulatory sandbox for telemedicine.

Industries: Healthcare, Life Sciences

Location: Singapore

Platform (<https://cloud.google.com/anidee/>),
BigQuery (<https://cloud.google.com/bigquery/>)

Google Cloud (<https://cloud.google.com/>)

Apigee API Platform

(<https://cloud.google.com/apigee/>)

Firebase Cloud Messaging

Cloud AI & Machine Learning

(<https://cloud.google.com/products/ai/>)

Google Workspace

Firebase (<https://firebase.google.com/>)

Firebase Authentication
(<https://firebase.google.com/products/auth/>)

Firebase Cloud Messaging

(<https://firebase.google.com/products/cloud-messaging/>)

Google Workspace

(<https://workspace.google.com/>)

Gmail

(<https://workspace.google.com/products/gmail/>)

Google Chat

(<https://workspace.google.com/products/chat/>)

COVID-19,
thanks to
Apigee for
API
management

Google Drive

(<https://workspace.google.com/products/drive/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

TensorFlow (<https://www.tensorflow.org/>)

The healthcare industry is digitally transforming in every area, from the way it delivers services to how it uses data to improve patient care. Why? Because the way people want to receive healthcare is changing, too. In one report, McKinsey found that

70% prefer online healthcare interactions

(<https://www.mckinsey.com/industries/healthcare-systems-and-services/our-insights/finding-the-future-of-care-provision-the-role-of-smart-hospitals>)

instead of on the phone or in person. Stepping in to meet these expectations are innovative service providers like Doctor Anywhere (<https://doctoranywhere.com/>).

Doctor Anywhere offers consumers the ease and convenience of healthcare services at their fingertips. Instead of spending time traveling to clinics or hospitals and waiting to be seen by a doctor, Doctor Anywhere allows consumers to access a range of services from consultation and claim management to prescription and wellness products, all in a single app.

The company wants to take health and wellness to the next level by empowering both consumers and medical providers through data. It runs a multicloud environment, anchored by a scalable big data platform on Google Cloud (<https://cloud.google.com/>).

The platform combines BigQuery (/bigquery) to speed up real-time analytics, Apigee (/apigee) to share data and services through secure APIs that enforce security rules related to authentication and authorization, and a range of Firebase (<https://firebase.google.com/>) services to understand user behavior.

“We want to be on the cutting edge of healthcare provision today, and that means adopting AI and machine learning technologies to access insights from data and use it to make our services more patient-centric. With its AI-first approach,

Google Cloud
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other cloud
providers.”

*—Rishik Bahri, Chief
Technology Officer,
Doctor Anywhere*

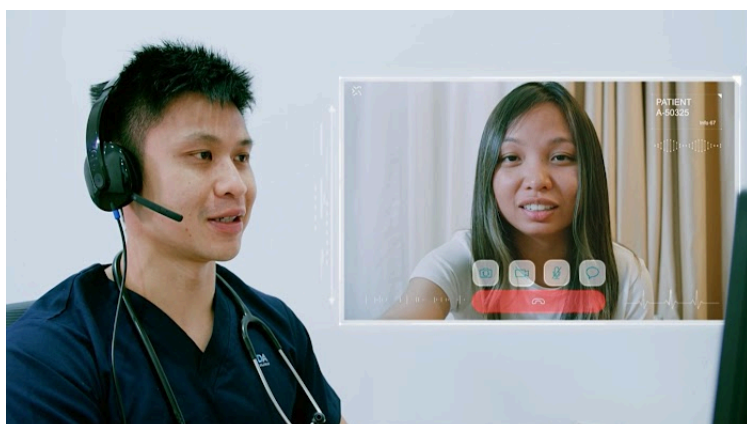
Many traditional health care providers are still reliant on paper-based medical records, a manual process that’s time-consuming for administrators and lacks transparency for consumers. Instead, Doctor Anywhere gives patients a holistic view of their healthcare history by providing consultation records, prescriptions, health screening reports, and medical certificates in the app.

Taking this one step further, Doctor Anywhere uses BigQuery to analyze patient data and other relevant data sources such as national health datasets to identify a patient’s risk category. This means that, instead of paying for costly treatments at the onset of illness, patients and physicians can focus on prevention to improve long-term health. And with BigQuery, doctors don’t need to be data scientists to gain real-time insights from health data.

“We want to be on the cutting edge of healthcare provision today,” says Rishik Bahri, Chief Technology Officer at Doctor Anywhere, “and that means adopting AI and machine learning technologies to

access insights from data and use it to make our services more patient-centric. With its AI-first approach, Google Cloud stands out from other cloud providers.”

He adds: “During the selection process, I was impressed by a medical use case where the presenter built a machine learning model on Google Cloud in under an hour. We’re exploring AI capabilities with [TensorFlow Enterprise](#) (/tensorflow-enterprise) and [Cloud AI](#) (/products/ai) to improve patient care by identifying macro trends, such as forecasting flu activity by looking at weather-related data.”



Building a data lake on BigQuery to eliminate data silos

Before switching to BigQuery on Google Cloud, Doctor Anywhere found it hard to integrate multiple data sources for in-depth analysis. To examine user behavior across channels, its data science team had to analyze data separately, from the in-clinic database and the online video consultation database, where electronic medical records of

patients are stored. It would run its analysis through a combination of SQL and Excel spreadsheets, then connect the insights from both to identify trends.

Today, Doctor Anywhere's data warehouse is completely modernized, with BigQuery helping to speed up the real-time analysis of 200 GB of data per month. And by using Apigee to manage its APIs, Doctor Anywhere can feed data from third-party providers directly into BigQuery, whether it's a lab delivering medical diagnoses or a government agency providing clinical data.

"BigQuery helps us take fragmented data, siloed across multiple platforms, and seamlessly combine it into a single, unified view," says Rishik. "Internally, it helps us run fast queries against all our data, allowing us to analyze our various product offerings to identify the ones that need to be improved," says Rishik. "On the client front, we are able to provide better personalized insight into a user's health based on anonymized data from various sources, to tailor personalized corporate wellness programs for them."

BigQuery also enables Doctor Anywhere to configure compute power for specific queries, allowing it to scale resources up to run thousands of queries a day, and then scale back down.

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*—Rishik Bahri, Chief
Technology Officer,
Doctor Anywhere*

Foster collaboration with the use of Apigee and APIs

A survey taken during COVID-19 reveals that 76% of consumers

(<https://www.mckinsey.com/industries/healthcare-systems-and-services/our-insights/telehealth-a-quarter-trillion-dollar-post-covid-19-reality>)

are interested in continuing with telehealth, or remote access to healthcare through technology, beyond the current climate. The survey found that healthcare providers agree, with 57% reported as more comfortable providing telehealth. It's a timely report for Doctor Anywhere, which is working to extend its digital services to new partners, from physicians to insurers, through the smart use of APIs and API management. Doctor Anywhere processes more than two million API calls each month from third-party apps and uses Apigee to secure access to backend services such as seeing a doctor on demand, using well-defined APIs.

"During COVID-19, we experienced an increase of more than 70% traffic on the Doctor Anywhere app," says Rishik. "Using Apigee for API management, we delivered frictionless access to patients and medical providers alike."

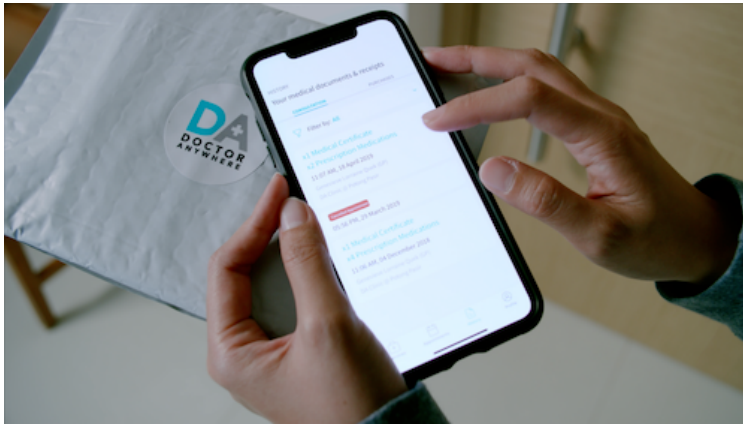
Before Apigee, many of its 100 APIs were exposed directly through Doctor Anywhere's backend application. Any API changes or enhancements meant a developer had to write code, test it, build it, and deploy in the QA environment before production. This whole process can take up to a month. Apigee acts as an API proxy between Doctor

Anywhere's client app and the backend server so developers can change service implementation, such as moving services to a new host, without affecting the public API.

"We tried out one API gateway, but it involved repetitive work and couldn't manage APIs at scale. In comparison, Apigee's flexible platform helps us quickly onboard new partners by assigning API access in minutes," says Rishik. "Instead of coding out access rights individually, our developers can focus on app improvements for our users."

Doctor Anywhere is expanding into Thailand and Vietnam where telehealth can significantly increase healthcare access for people living in remote areas. Patients can video consult a local certified doctor on the Doctor Anywhere app, without traveling long distances for quality medical care. Operating in different countries, Doctor Anywhere wants to make sure that app content is relevant to the user depending on their location.

"Instead of writing code in the application, we use Apigee to call the external web servers, which provides the country of visitors based on their IP addresses," says Rishik. "That information is then integrated into our backend APIs to connect patients to the locally available doctors in our database."



Creating a richer app experience with Firebase to engage users

Behind the scenes, Doctor Anywhere saves time by using Firebase to jumpstart its mobile application with pre-made components, from authentication to push notification. [Firebase Authentication](#)

(<https://firebase.google.com/docs/auth>) allows a patient to register on a Doctor Anywhere app using third-party apps such as their insurance provider's app or personal social platforms.

After logging in, the patient can book an appointment with a wide range of healthcare professionals, from general practitioners to mental health experts. The request is sent from the server through [Firebase Cloud Messaging](#)

(<https://firebase.google.com/products/cloud-messaging>) to the [DA - Provider](#)

(https://play.google.com/store/apps/details?id=com.doctoranywhere.provider&hl=en_SG)

app on the doctor's mobile device so the doctor can accept or reject the video consultation, based on their availability.

Firestore In-App Messaging

(<https://firebase.google.com/products/in-app-messaging>)

improves patient communication with appointment reminders and post-treatment follow-up. The marketing team at Doctor Anywhere also uses In-App Messaging to engage users with rich content. For example, during the COVID-19 pandemic, the team promoted a free mask and Vitamin C giveaway by displaying the DA Care Kit campaign image in the app, without writing any code. Additionally, Google Analytics for Firebase

(<https://firebase.google.com/products/analytics>)

measures user engagement so the marketing team can improve campaign performance.

"With the help of In-App messaging, our marketing team doesn't need coding experience to launch interactive pop-ups in the application in minutes," says Rishik. "They can review campaign results from the Firebase console or Google Analytics to see how users interact with the campaign and create funnels to increase conversion."

The engineering team also uses analytics to better understand and meet the needs of its users. "Based on past trends, we identified that many young people may not be covered under outpatient medical insurance," says Rishik. "So we launched the DA Health Protect program, a subscription module which allows users to access our online and offline clinics at a flat \$13 consult fee for an annual subscription fee of \$12, to help ease medical costs."

“By combining BigQuery and machine learning, we hope to uncover links between demographic factors and medical conditions and share personalized health product recommendations based on a patient’s risk profile. Through data insights and machine learning on Google Cloud, we believe we can transform healthcare, helping people

stay healthy and
happy.”

—Rishik Bahri, Chief
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Improving communication with Google Workspace to accelerate innovation

Effective collaboration is integral to the success of Doctor Anywhere. Members from different departments, such as marketing, sales, and engineering, work together in cross-functional teams to create new products and features. Doctor Anywhere uses Google Workspace (<https://workspace.google.com/>) products, such as Google Chat (<https://workspace.google.com/products/chat/>), Meet (<https://workspace.google.com/products/meet/>), and Drive (<https://workspace.google.com/products/drive/>) to improve communication across offices in Singapore, India, Vietnam, and Thailand.

“During the COVID-19 pandemic, our developers in India and Singapore conducted daily meetings on Meet while working remotely,” says Rishik. “Third-party integration with Jira makes it easier for the help desk to assign tasks to developers within Chat so they can fix bugs quickly for better user experience.”

Doctor Anywhere works with partners in highly regulated healthcare and insurance industries. Using access control for Chat provides partners with a peace of mind that Doctor Anywhere complies with their specific data-sharing guidelines such as the International Organization for Standardization and Monetary Authority of Singapore.

“Google Drive improves our collaboration by giving us a common document library to securely share and access documents within the company, regardless of the employee’s location,” says Rishik. “By using keyword searches, we can easily find relevant documents and information stored in Drive as well Chat and Gmail (<https://workspace.google.com/products/gmail/>).”

The team also relies heavily on Chat and Meet to stay in touch with its teammates in India, Vietnam, and Thailand. “We regularly use the shared drive feature on Drive to collaborate and share documents with team members securely and seamlessly.”

Powering up digital healthcare with Google Cloud AI

Moving forward, Doctor Anywhere wants to provide healthcare that’s even more patient-centric, with the help of machine learning. “By combining BigQuery and machine learning, we hope to uncover links between demographic factors and medical conditions and share personalized health product recommendations based on a patient’s risk profile,” says Rishik. “Through data insights and machine learning on Google Cloud, we believe we can

transform healthcare, helping people stay healthy
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